

LEAVING CERTIFICATE BIOLOGY · 40% OF THE FINAL GRADE

Leaving Certificate Biology - Investigation Brief 2027

Biology in Practice Investigation · issued by the State Examinations Commission,
January 2026 (recreated for this demo)

1. To the candidate

The State Examinations Commission (SEC) publishes a brief each year for the Biology in Practice Investigation, the 40% Additional Assessment Component of Leaving Certificate Biology. This brief is for candidates examining in 2027 and was issued in January 2026, in Term 2 of Year 1.

The brief gives the context and stimulus for your investigation. You will complete scientific research on an issue related to the brief, design and conduct an experiment to generate your own primary data, and develop an evidence-based argument in response to the brief. The investigation takes approximately 20 hours across the course, and it is designed to be part of your ordinary biology classwork, not separate from it.

2. The stimulus: three investigation topics

Choose ONE of the three topics below as the starting point for your research and experiment. Each is open-ended: you decide the particular issue, the research question and the experiment.

Topic 1 - Membranes

Every cell is enclosed by a membrane – a partially permeable barrier that controls what enters and leaves. Membranes are made of molecules that are sensitive to their surroundings: heat, alcohol and acids can disrupt them, changing how permeable they become.

Everyday starting points: why beetroot stains the cooking water when it is boiled; why alcohol stings on a cut; how the inside of an egg models a cell; what happens to thawed frozen strawberries; why some plants wilt in heat but recover when watered.

Topic 2 - Osmosis

Water moves across partially permeable membranes from where water is more concentrated to where it is less concentrated. This movement – osmosis – matters to every living thing, all the time.

Everyday starting points: why salting or brining draws water out of food; why salad wilts and crisp lettuce revives in water; what sports drinks are designed to do; why concentrating sugar preserves jam; how plant cells become firm or limp.

Topic 3 - Food preservation

Food spoils when microorganisms grow on it and enzymes inside it keep working. Preservation methods – cold, heat, salt, sugar, acid, drying, vacuum and sealing – all work by slowing or stopping that growth and activity.

Everyday starting points: why milk lasts longer in the fridge than on the counter; what brine does to bacon or cabbage; why jam keeps once opened only if refrigerated; how yeast behaves in bread-making and why salt is measured carefully; why frozen peas taste fresh months later.

3. The task

Having chosen a topic, complete two connected pieces of work:

- Scientific research on an issue related to your chosen topic: draw on secondary sources, develop a research question, review and summarise evidence from different viewpoints, and keep a record of findings, download dates and references in your investigative log.
- An experiment related to your chosen topic: pose a testable hypothesis, plan and design the experiment, conduct it under your teacher's supervision in the laboratory or a field setting, gather primary data, analyse it, and form justifiable conclusions.

4. Requirements

Keep an investigative log throughout – a personal record of your approach, decisions, data and reflections. The log is not submitted to the SEC, but it is where your report comes from. The investigation is your own individual work; you may get help from peers only to handle equipment safely, and the data you use must be your own.

Students should reflect on difficulties experienced during planning, record how they overcame them, and treat unanticipated outcomes as a valid and important part of the scientific process.

- Approximately 20 hours in total across the stages: initial response (1-2 hours), background research (2-3 hours), designing and planning (2-3 hours), conducting (depends on the brief), analysis and conclusions (1-2 hours), finalising the report (up to 4 hours).
- Reference every source that is not your own – including any AI tools: the tool's name, the date, how it was used, and the shareable chat URL or the prompt used.

- The report is submitted in Year 2 in a digital format prescribed by the SEC; the accompanying instructions set the word count, number of images, required structure and section headings.

5. Timeline

This brief issued in January 2026 (Term 2 of Year 1). Stages are completed when they best fit teaching and learning across 5th year – they do not need to run in one continuous block. All work must be concluded by the date set by the SEC, and the report is submitted for marking in Year 2.

6. Authenticity and the use of AI

All work submitted for assessment must be your own. Submitting work not entirely completed by you is a significant breach of regulations and may lead to penalties, up to and including the withholding of results. Material generated by AI software must be acknowledged in your references. Direct copying from any source, including AI-generated material, is not permitted.